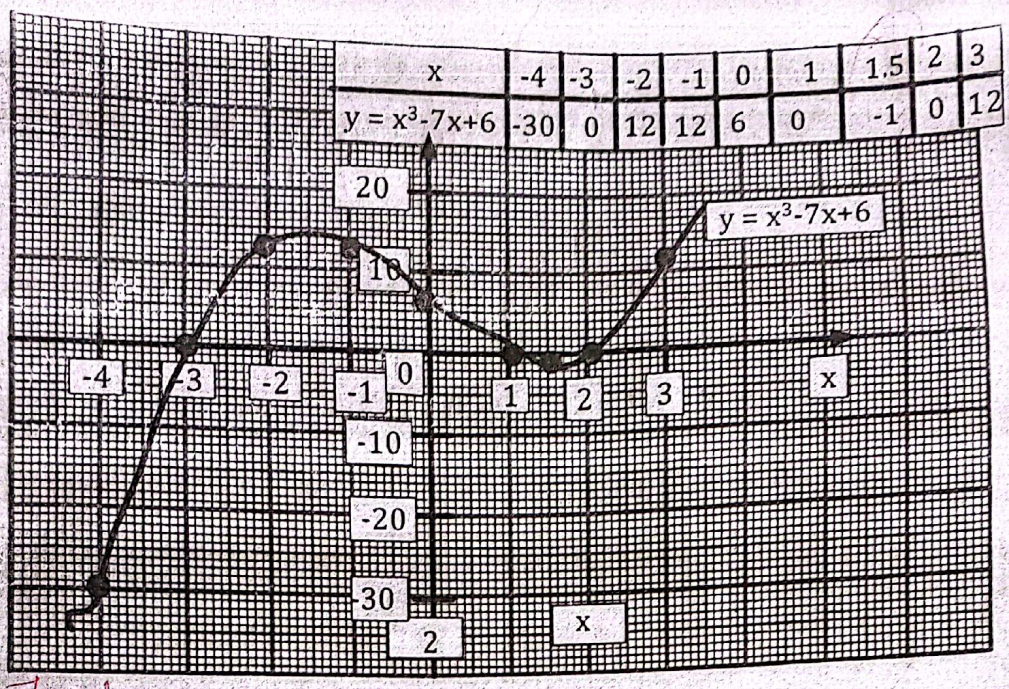
MARKING GUIDE 2R2236

Item 1

x	-4	-3	-2	-1	0	1	1.5	2	3
$y = x^3-7x+6$	-30	0	12	12	6	0	-1	0	12



T_1-1 Table
M_1-1 out put atleast 3
M_1-1 scaling
M_1-1 labelling axes atleast one
M_1-1 For plotting
M_1-1 for curve
B_1-1 for points 07

The values of x are -3, 1 and 2

T_1-1 for identify trapezium rule (h)

$$h = \frac{3-2}{5} = 0.2$$

X	$y = x^3 - 7x + 6$	
2	0.000	
2.2		1.248 ~~3.024~~
2.4		2.424 3.024
2.6		3.576 5.376
2.8		4.752 8.352
3.0	6.000 ~~12.000~~	
Sum	6.0000 ~~12.0000~~	12.000 ~~18.000~~

for correct (h) M_1-1, correct values of x M_1-1

$$\int_2^3 (x^3 - 7x + 6)\,dx \approx \frac{1}{2} x 0.2[\overset{12}{6} + 2(\overset{18}{\cancel{12}})]$$

$$\approx \cancel{3.00}\ 4.860\ sq\ units$$ M_1-1 for all correct values of y, M_1-1 correct sub in FP, D_1-1 07

M_1-1 for out put

Since ~~3~~ 4.874 ~~< 4~~ enough, The site is too small for the proposed public washroom

T_1-1 for errors' concept

$r = 80.45\ cm, e_r = \pm 0.005$

$h = 120.2\ cm, e_h = \pm 0.05$

M_1-1 Sub formula

$V = \pi r^2 h.$

$V = \pi(80.45)^2 x 120.2$

M_1-1 for out put

$V = 777958.7405\pi\ cm^3$

$= 2444029.463958\ cm^3$ (calc)

$2444301.844286\ cm^3$ (for 22/7)

1

$V_{max} = \pi r^2_{max} x h_{max} = \pi(80.455)^2 x 120.25$

$V_{max} = 778379.0948\pi\ cm^3$ 2446334.297854cm³ for 22/7

2445,350.045931 6cm³ (cal)

$V_{min} = \pi r^2_{min} x h_{min} = \pi(80.445)^2 x 120.15$

$V_{min} = 777538.4727\pi\ cm^3$ 2443692.342783 2cm³ for 22/7

2442,709.153717 7cm³ (cal)

$|e_{Vol}| = \frac{1}{2}(V_{max} - V_{min}) = \frac{1}{2}(778379.0948\pi - 777538.4727\pi)$

$|e_{Vol}| = 420.311\pi$ 1320.977428571 4cm³ for 22/7

1320.445949823cm³ (cal)

$P.E_{Vol} = \frac{|e_{Vol}|}{Vol} x100 = \frac{420.311\pi}{777958.7405\pi} x100$

$P.E_{Vol} = 0.054\%$

Since $0.054\% < 1.0\%$, the pavers will be accepted

M1 − 1 for max out

M1 − 1

M1 − 1 for correct output

M1 − 1 for sub

M1 − 1 for out put

M1 − 1 for sub

M1 − 1 for out put

D1 − 1

24

ALT:

Volume; $V = \pi r^2 h$. T1 − 1

For products and powers, $\frac{\Delta V}{V} = 2\frac{\Delta r}{r} + \frac{\Delta h}{h} = 2\left(\frac{0.005}{80.45}\right) + \frac{0.05}{120.2}$ M1−1 M1−1 M1−1 M1 − for addition

$\frac{\Delta V}{V} = 0.00054.$ M1−1 M1−1 M1−1

Percentage error $= 0.00054 \times 100 = 0.054\%$.

Since $0.054 < 1$, the company should proceed with purchasing the culverts. D1 − 1

10

Item 2

X	P(X=x)	$xP(X = x)$	$x^2 P(X = x)$
0	0.02	0	0
20	0.16	3.2	64
40	0.22	8.8	352
60	0.26	15.6	936
80	0.20	16.0	1280
100	0.14	14.0	1400
		$E(x) = 57.2576$	$E(x^2) = 4032$

T1 − 1 for xp(x=x)

T2 − 1 for x^2P(X=x)

5

Expected number of pieces sold per day $= 57.6$ M2−1

M2 − 1 For out put

M2 − 1 For out put

Variance: $Var(X) = E(X^2) - [E(X)]^2$

$Var(X) = 4032 - (57.6)^2 = 714.24$

M2 − 1 for var

Standard deviation: $\sigma = \sqrt{714.24} = 26.7253$

M2 − 1 for SD

07

Expected daily profit

Each chicken costs UGX 3,000.

Daily purchasing cost: $100 \times 3000 = 300{,}000$ UGX
Revenue from selling X pieces: $6000X$
Profit: $P = 6000X - 300000$
Expected profit: $E(P) = 6000E(X) - 300000$

$$= 6000(57.6) - 300000$$

$$= 45{,}600$$

$$E(P) = \text{UGX } 45{,}600$$

$M_2 - 1$ for cost price
$M_2 - 1$ for profit fun
$M_2 - 1$ for Property
$M_2 - 1$ for out
or 48,000 in 58
$D_2 - 1$
25

Since $45{,}600 < 50{,}000$, the vendor **should apply for the loan** according to her plan.

Alternatively

(a) $Profit = S.P - C.P$

0 pieces, $Profit$ = 0x6000−100x3000 = −300,000

20 pieces, $Profit$ = 20x6000−100x3000 = −180,000

40 pieces, $Profit$ = 40x6000−100x3000 = −60,000

60 pieces, $Profit$ = 60x6000−100x3000 = 60,000

80 pieces, $Profit$ = 80x6000−100x3000 = 180,000

100 pieces, $Profit$ = 100x6000−100x3000 = 300,000

$M_2 - 1$ for any correct expression put
$M_2 - 1$ for all correct values

Y	P(Y=y)	$yP(Y = y)$
-300,000	0.02	-6000
-180,000	0.16	-28,800
-60,000	0.22	-13,200
60,000	0.26	15,600
180,000	0.20	36,000
300,000	0.14	42,000
		$E(Y) = 45{,}600$

$M_2 - 1$ for all values of $yP(Y=y)$
$M_2 - 1$ for out put

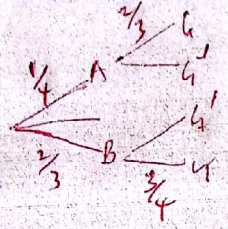
Since $45{,}600 < 50{,}000$, the vendor **should apply for the loan** according to her plan.
$D_2 - 1$

Given $P(A) = \frac{1}{4}$, $P(B) = \frac{3}{4}$, $P(G \mid A) = \frac{2}{3}$, $P(G \mid B) = \frac{1}{3}$

Using the law of total probability, $P(G) = P(A)P(G \mid A) + P(B)P(G \mid B)$

$T_2 - 1$ for Conditional probability

CamScanner



$$= \frac{1}{4}\cdot\frac{2}{3} + \frac{3}{4}\cdot\frac{1}{3}$$

$$= \frac{5}{12} \approx 0.4167$$

Since $\frac{5}{12} = 0.4167 > \frac{2}{5} = 0.4$, the vendor **does not need to change suppliers.**

Let: $X \sim \text{Bin}\left(5, \frac{1}{4}\right)$

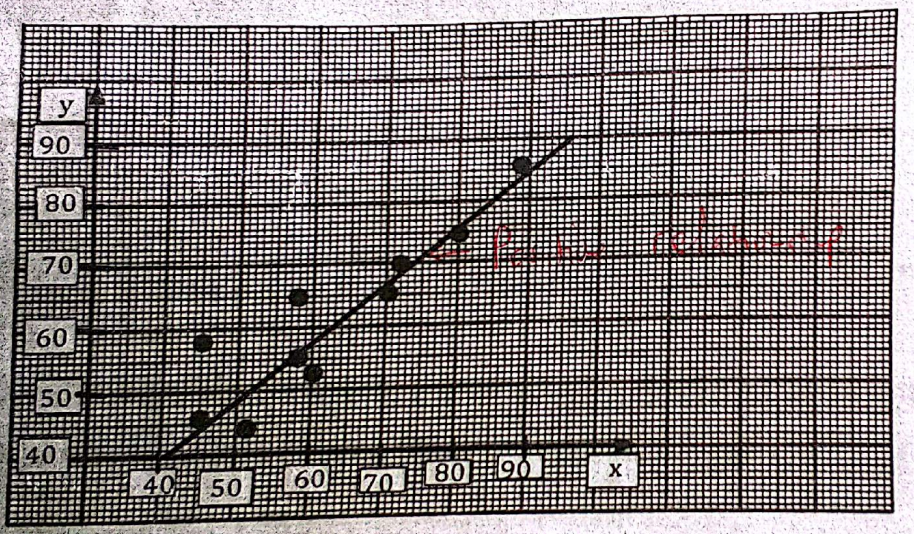
We require $P(X = 3)$

Using the binomial formula, $P(X = 3) = \binom{5}{3}\left(\frac{1}{4}\right)^3\left(\frac{3}{4}\right)^2$

$$= 10\left(\frac{1}{64}\right)\left(\frac{9}{16}\right)$$

$$= \frac{90}{1024} \text{ Or } \frac{45}{512} \approx 0.0879$$

Item 3



	R_x	R_y	d^2
7	4	4	0
6	5	8	9
4.5	6.5	7	0.25
1.5	8.5	9	0.25
9	2	2	0
10	1	1	0
4.5	8.5	6	6.25
8	3	3	0
	6.5	4.5	4
1	10	10	0
		Σd^2 =	19.75

$$\rho = 1 - \frac{6\Sigma d^2}{n(n^2-1)} = 1 - \frac{6\times 19.75}{10099}$$

$$\rho = 0.8182$$

Its significant at 5%

C.B	x	f	fx	fx^2	$f.d$
35 – 45	40	4	160	6400	0.4
45 - 50	47.5	13	617.5	29331.25	2.6
50 - 60	55	25	1375	75,625	2.5
60 - 65	62.5	30	1875	117187.5	6.0
65 - 75	70	20	1400	98000	2.0
75 - 80	77.5	15	1162.5	90093.75	3.0
80 - 95	87.5	3	262.5	22968.75	0.6
		$\sum f =$ 110	$\sum fx =$ 6852.5	$\sum fx^2 =$ 439606.25	

$T_2 - 1$ for fx
$T_3 - 1$ for fx^2
$T_3 - 1$ for $f.d$

Mean $= \frac{\Sigma fx}{\Sigma f} = \frac{6852.5}{110} = 62.2955$

$M_3 - 1$ for sub, $M_3 - 1$ for output

Standard deviation $= \sqrt{\frac{\Sigma fx^2}{\Sigma fx} - (\bar{x})^2} = \sqrt{\frac{439606.25}{110} - (62.2955)^2}$

$M_2 - 1$ for sub
$M_3 - 1$ for output

$\sigma = 10.7562$

Mode$= L_b + \left(\frac{\Delta_1}{\Delta_1+\Delta_2}\right)c = 60 + \left(\frac{6-2.5}{6-2.5+6-2.0}\right)5$

$M_2 - 1$ for sub
$M_3 - 1$ for output

$Mode = 62.3333$

$D_3 - 1$

Since 62.3333 > 55, the administration **will not organize a student meeting**

$T_3 - 1$, $M_3 - 1$ attempts to standardizing

$P(X > 85) = P\left(Z > \frac{(85 - 62.2955)}{10.7562}\right)$

$= P(Z > 2.111)$

$= 0.0174$ 0.01739 (cal)

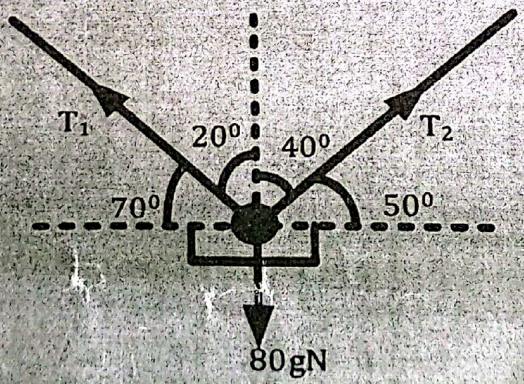
$M_3 - 1$

$M_3 - 1$

25

Number=0.0174x110=1.9≈ 2 students

Item 4



$T_4 - 1$ correct force diagram

$T_4 - 1$ angles correctly indicated

T_4 – for force diagram

$$\frac{T_1}{\sin 140^\circ} = \frac{784}{\sin 60^\circ}$$

$$T_1 = 581.9061N$$

$$\frac{T_2}{\sin 160^\circ} = \frac{784}{\sin 60^\circ}$$

$$T_2 = 309.6258N$$

Since both tensions exceed **300 N, both ropes should be replaced**

Alternatively: Horizontal equilibrium: $T_1 \cos 70^\circ = T_2 \cos 50^\circ$

$$T_1 = \frac{\cos 50^\circ}{\cos 70^\circ} T_2 = 1.879\, T_2$$

Vertical equilibrium: $T_1 \sin 70^\circ + T_2 \sin 50^\circ = 784$

Substitute $T_1 = 1.879T_2$:

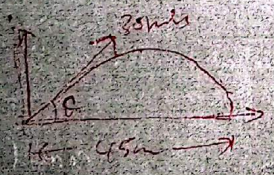
$$1.879T_2(\sin 70^\circ) + T_2(\sin 50^\circ) = 784$$
$$1.879(0.9397)T_2 + 0.7660T_2 = 784$$
$$2.5316T_2 = 784$$
$$T_2 = 309.6258N$$

Hence: $T_1 = 1.879(309.7)$

$$T_1 = 581.9061N$$

Since both tensions exceed **300 N, both ropes should be replaced.**

Projectile formula: $R = \frac{u^2 \sin 2\theta}{g}$



$$45 = \frac{30^2 \sin 2\theta}{9.8}$$
$$\sin 2\theta = 0.49$$
$$2\theta = \sin^{-1}(0.49)$$
$$2\theta = 29.34^\circ, 180^\circ - 29.34^\circ$$

Therefore: $\theta = 14.67^\circ$ *Or* $\theta = 75.33^\circ$

Thus the football can be kicked at either: *14.7° or 75.3° to land* ***45 m*** *away.*



$$S = ut + \frac{1}{2}at^2$$

For AB: $500 = u60 + \frac{1}{2}a(60)^2$

$$15u + 900a = 125 \ldots\ldots\ldots\ldots\ldots\ldots\ldots (1)$$

For AC: $1000 = u240 + \frac{1}{2}a(240)^2$

M4 – 1 for Sub
M4 – 1 For output
M4 – 1 For Sub
M4 – 1 For output
D4 – 1 OF

M4 – 1 for Resv. horizontally or vertically.
M4 – 1 for Vertical
M4 – 1 for solving 2 eqtns

M4 – 1 for T_2
M4 – 1 for T_1
D4 – 1

T4 – 1 for projectile motion
M4 – 1 substituting
OF

M4 – 1, M4 – 1 for 2θ, for θ

T4 – 1 for linear motion
M4 – 1 for AB
M4 – 1 for AC

$$3u + 720a = 12.5 \ldots\ldots\ldots\ldots\ldots\ldots\ldots (2)$$

$(1) - 5(2)$: $-2700a = 62.5$

$$a = -0.0231 \text{ m s}^{-2} \; Or \; -\frac{5}{216} \text{ m s}^{-2}$$

$$u = \frac{12.5 - 720x(-0.0231)}{3}$$

$$u = 9.7222 \text{ m s}^{-1} \; Or \; \frac{175}{18} \text{ m s}^{-1}$$

Distance travelled before stopping: $v^2 = u^2 + 2as$

$$0 = \left(\frac{175}{18}\right)^2 + 2\left(-\frac{5}{216}\right)s$$

$$s = 10{,}335.9375 \text{ m}$$

Extra distance= 10,335.9375 -1000=9,335.9375m

Item 5

At 12:00

$$\underset{\sim}{R}_{T(t=1)} = \begin{pmatrix} 2 \\ -3 \\ 4 \end{pmatrix} + 1\begin{pmatrix} 2 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 4 \\ -1 \\ 7 \end{pmatrix} km$$

$$_T\underset{\sim}{R}_{B(t=t)} = \underset{\sim}{R}_{T(t=t)} - \underset{\sim}{R}_{B(t=t)} = \begin{pmatrix} 4 \\ -1 \\ 7 \end{pmatrix} + t\begin{pmatrix} 2 \\ 2 \\ 3 \end{pmatrix} - \begin{bmatrix} 7 \\ 5 \\ -2 \end{bmatrix} + t\begin{pmatrix} -3 \\ -8 \\ 18 \end{pmatrix}$$

$$_T\underset{\sim}{R}_{B(t=t)} = \begin{pmatrix} -3 + 5t \\ -6 + 10t \\ 9 - 15t \end{pmatrix} km$$

For collision, $\begin{pmatrix} -3 + 5t \\ -6 + 10t \\ 9 - 15t \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$

i; $-3 + 5t = 0 \Rightarrow t = 0.6$ hours

j; $-6 + 10t = 0 \Rightarrow t = 0.6$ *hours*

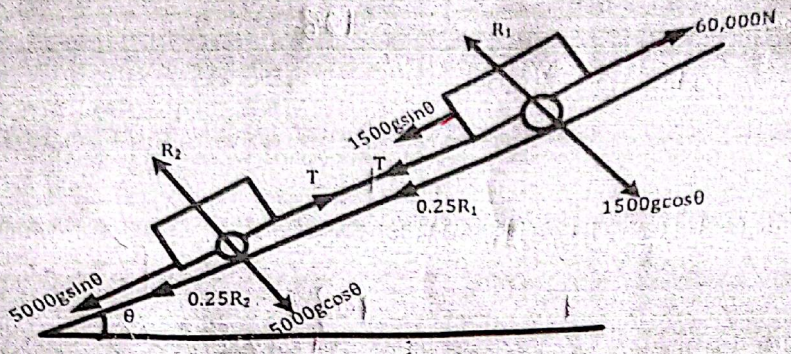
k; $9 - 15t = 0 \Rightarrow t = 0.6$ *hours*

Since $t_i = t_j = t_k = 0.6$ *hours*, then collision occurs at

$= 12{:}00\text{pm} + 0.6(60)$ minutes

$= 12{:}36$ pm

$$\underset{\sim}{R}_{T(t=36)} = \begin{pmatrix} 4 \\ -1 \\ 7 \end{pmatrix} + 0.6\begin{pmatrix} 2 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 5.2 \\ 0.2 \\ 8.8 \end{pmatrix} km$$



$$\sin\theta = \frac{3}{5}, \cos\theta = \frac{4}{5}$$

For 1500kg: $60{,}000 - (1500gsin\theta + 0.25R_1 + T) = 1{,}500a$

$$60{,}000 - (8820 + 2940 + T) = 1{,}500a$$

$$48{,}240 - T = 1{,}500a \ldots\ldots\ldots\ldots (1)$$

For 5000kg: $T - (5000gsin\theta + 0.25R_2) = 5000a$

$$T - (29{,}400 + 9{,}800) = 5{,}000a$$

$$T - 39200 = 5000a \ldots\ldots\ldots\ldots (2)$$

$(1) + (ii)$: $a = 1.3908\, ms^{-2}$

$$T = 5{,}000(1.3908) + 39{,}200$$

$$T = 46{,}154N$$

TB -1 correct for force diagram

MS -1 for tansin

MS -1 for F=ma

MS -1 for applying F=ma

M.S -1

MS -1

MS -1

20